PhishGuard

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Company Name: BharatCares(CSRBOX Group)

Bootcamp Duration: 2 Weeks

Mode: Online

Topic: Cybersecurity

Introduction

PhishGuard: Phishing Email Detection and Analysis Tool is a cybersecurity project designed to help users identify potentially fraudulent or suspicious emails. Phishing is a common cyberattack in which attackers trick people into revealing sensitive information, such as passwords, banking details, or personal data. PhishGuard analyzes the content of an email for warning signs such as suspicious links, urgent language, and requests for confidential information. It provides a risk score, classifies the potential threat, and explains the reasons behind its analysis. The project aims to improve users' awareness of phishing attacks and encourage safer email practices through a simple, user-friendly interface.

Problem Statement

Phishing attacks are a common cybersecurity threat in which attackers use deceptive emails to trick users into revealing sensitive information, clicking malicious links, or performing unsafe actions. Many users struggle to identify these emails because they often appear genuine and create a sense of urgency. PhishGuard addresses this problem by analyzing email text for common phishing warning signs, assigning a risk score, and explaining suspicious indicators. It aims to help users recognize potentially harmful emails and make safer decisions. However, the tool uses basic rule-based detection and cannot guarantee the identification of every phishing attack.

Objectives

1. To analyze submitted email text for at least 8 common suspicious phrases and patterns.
2. To detect URLs and identify requests for sensitive information, such as passwords and OTPs.
3. To generate a risk score from 0 to 100 and classify emails into Low, Medium, or High risk.
4. To display detected warning signs and provide at least 3 relevant safety recommendations.
5. To store scan results during the current session and display the 10 most recent scans.
6. To enable users to export scan history in CSV format for further review.
7. To provide Dark, Light, and High Contrast themes, along with adjustable text size and read-aloud accessibility features.

Tools and Technologies

- **Python:** Main programming language used to develop the application.
- **IBM Bob:** AI-assisted coding tool used to support development.
- **Streamlit:** Used to create the interactive web interface.
- **Regular Expressions (re):** Used to detect suspicious phrases and URLs in email text.
- **CSV and io modules:** Used to export scan history in CSV format.

- **HTML and CSS:** Used to style the interface and implement theme and accessibility features.
- **JavaScript:** Used to implement browser-based text-to-speech functionality.

Working/ Methodology

PhishGuard follows a rule-based approach to analyze email content and identify potential phishing indicators.

1. Email Input**:** The user pastes email text into the Email Analyzer or loads a sample email.
2. Text Analysis**:** Python uses predefined rules and regular expressions to check for suspicious phrases, URLs, and requests for sensitive information.
3. Risk Scoring: The application assigns points for detected warning signs and calculates a risk score between 0 and 100.
4. Threat Classification**:** Based on the score, the email is classified as Low (below 30), Medium (30–59), or High (60–100) risk.
5. Results and Recommendations: The application displays the risk score, detected warning signs, and safety recommendations.
6. History and Export**:** Scan results are stored in the current session, displayed on the Security Dashboard, and can be downloaded as a CSV file.

The application also provides accessibility features, including multiple themes, adjustable text size, and text-to-speech support.

Implementation

The PhishGuard application is implemented in Python using Streamlit. Its main code components are:

1. Application Setup: Configures the page layout, title, session state, and interface styling.
2. **Email Analysis Function:** The analyze_email() function uses predefined rules and regular expressions to detect suspicious phrases, URLs, and requests for sensitive information.

3. **Risk Assessment:** Calculates a risk score from 0 to 100 and assigns a Low, Medium, or High risk classification.
4. **Email Analyzer:** Accepts email text, loads sample emails, and displays analysis results with warning signs and safety recommendations.
5. **Security Dashboard:** Displays scan statistics, risk distribution, recent results, and scan history, with CSV export functionality.
6. **Safety Guide:** Provides guidance on recognizing suspicious senders, urgent requests, unsafe links, and attempts to obtain confidential information.
7. **Accessibility Features:** Supports Dark, Light, and High Contrast themes, adjustable text size, and browser-based text-to-speech.

The application uses session state to maintain scan history temporarily during the current Streamlit session. It does not use a permanent database or a machine-learning model.

Output

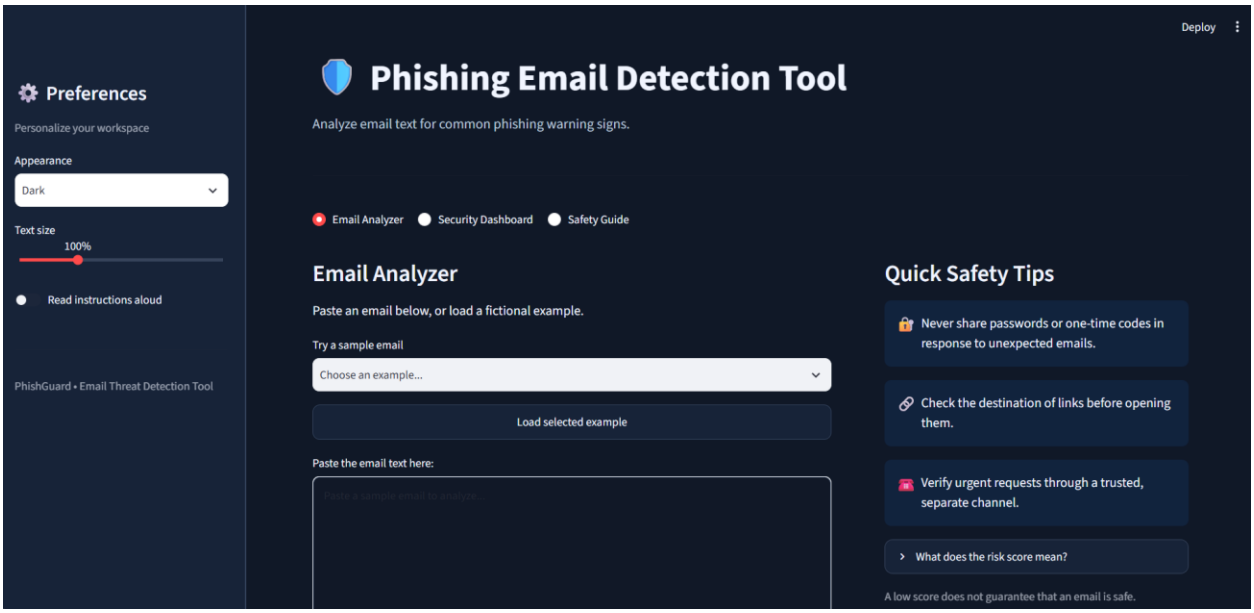


Figure 1 home page

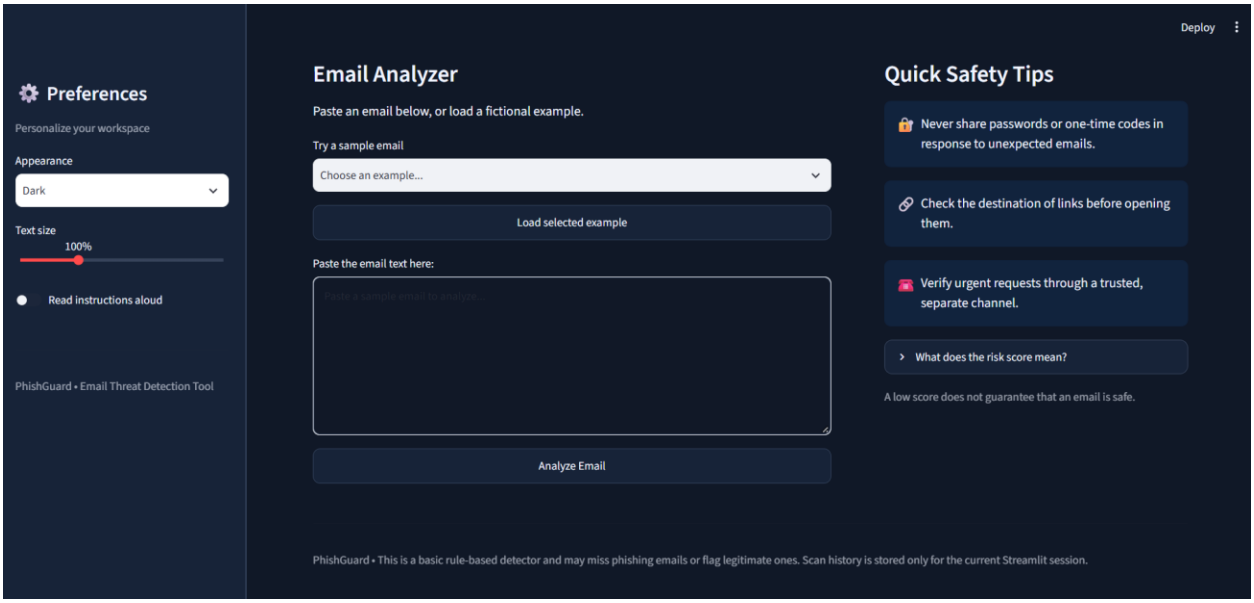


Figure 2 home page

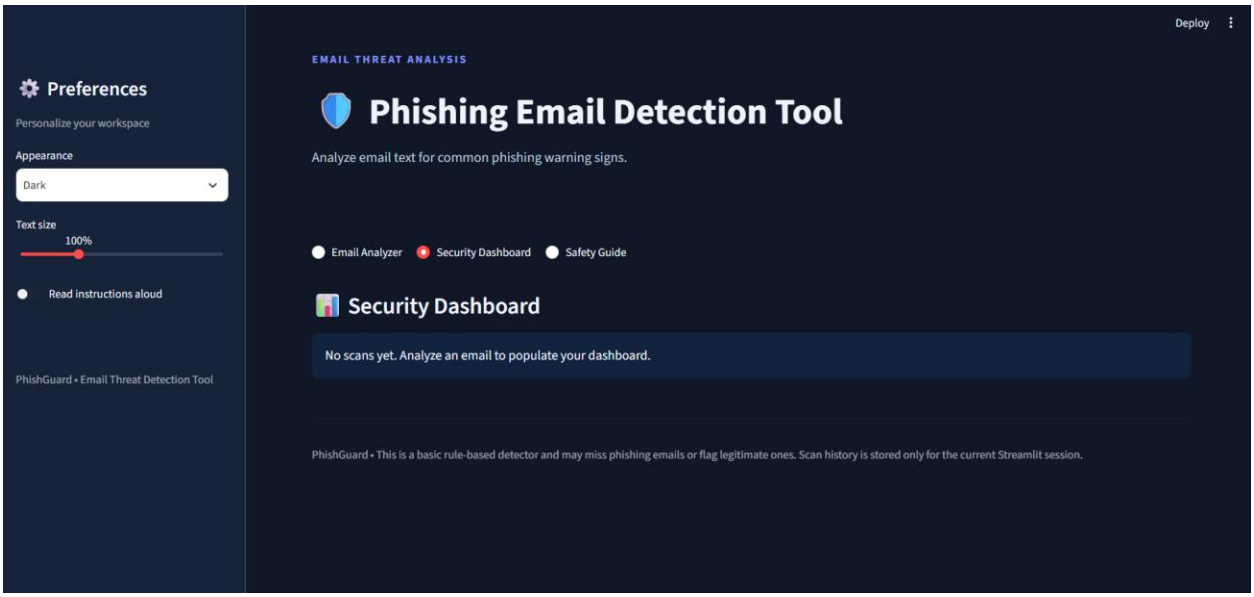


Figure 3 Security Dashboard



Figure 4 Safety Guide

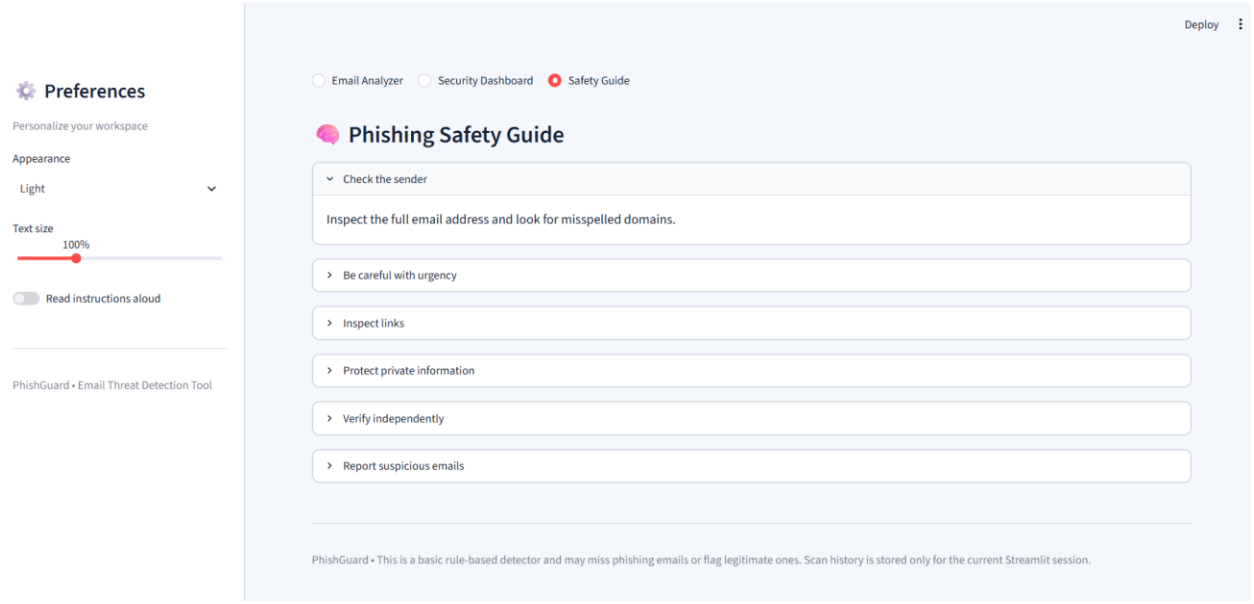


Figure 5 Light mode

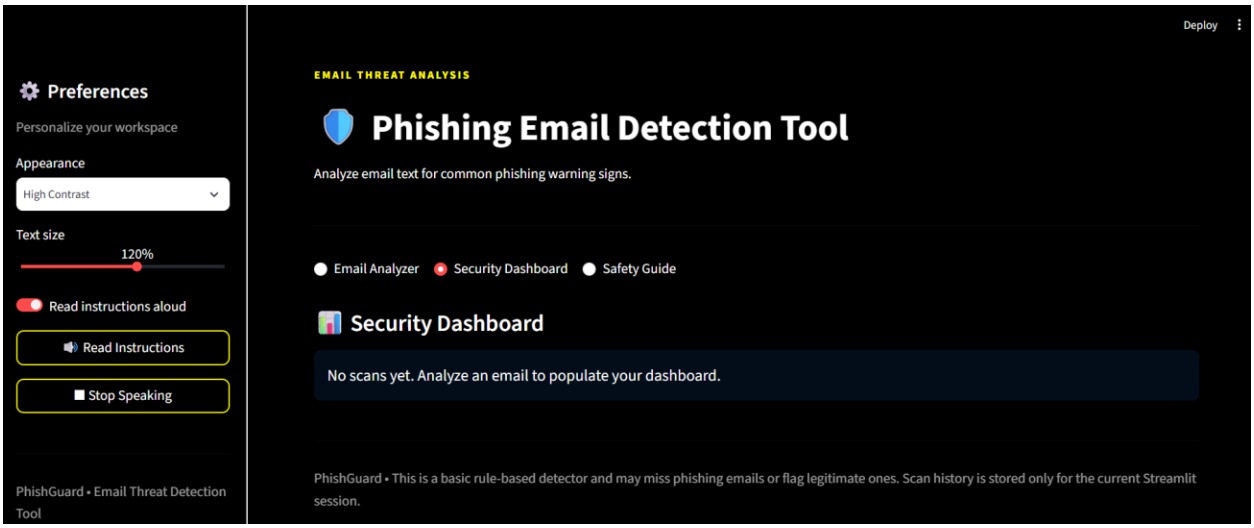


Figure 6 High Contrast mode

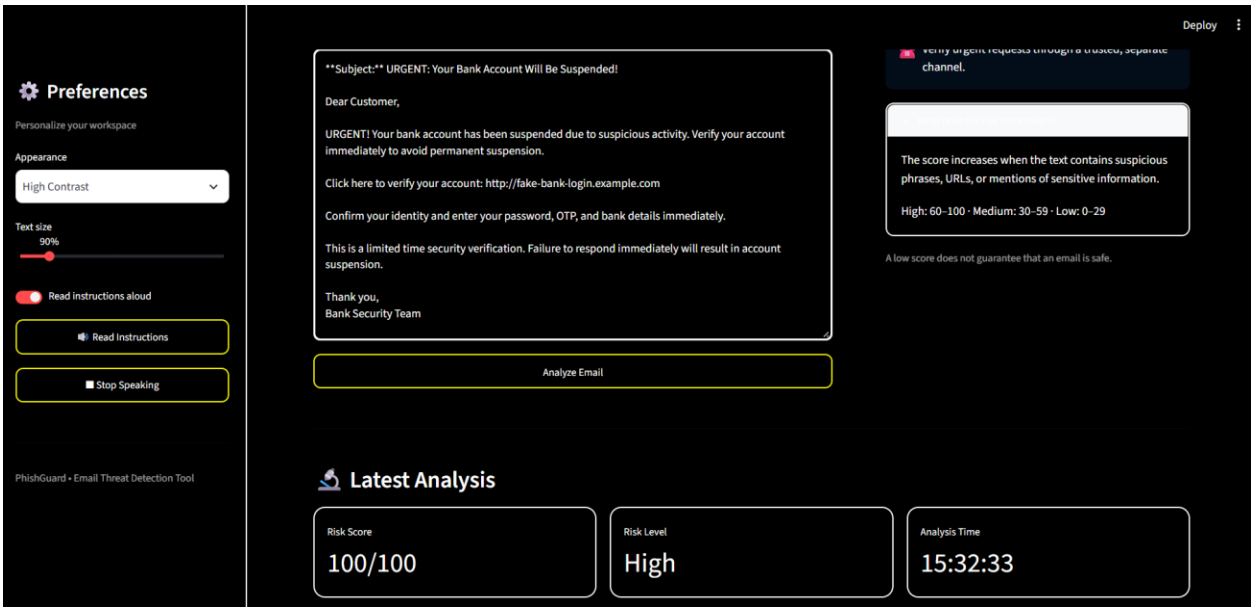


Figure 7 Example of phishing email

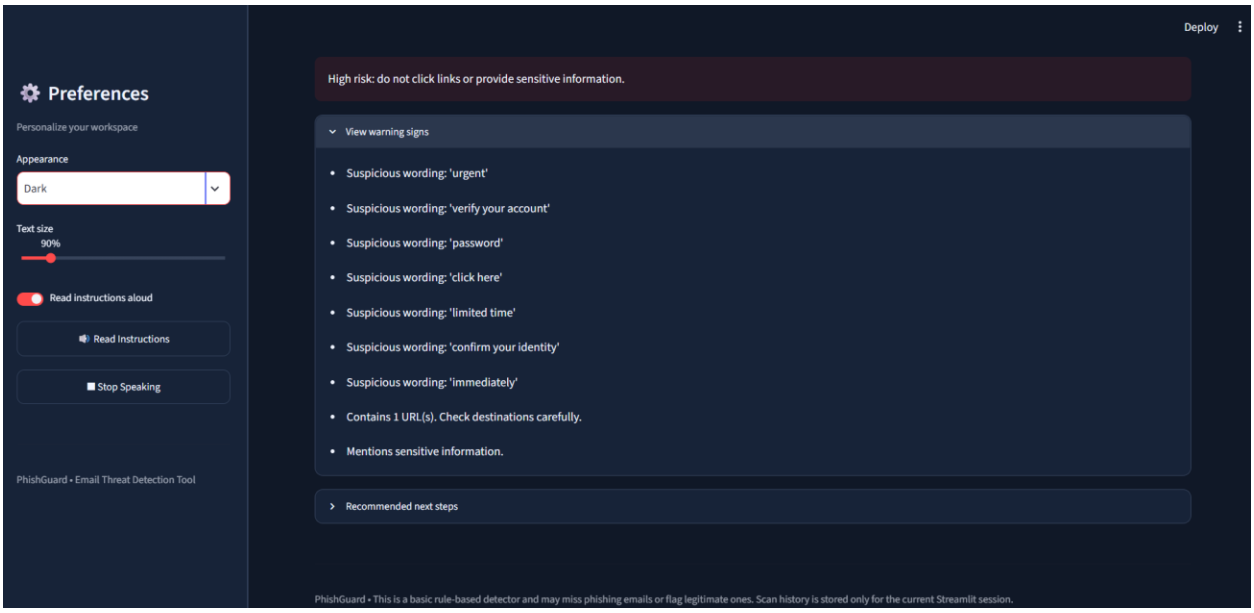


Figure 8 Result

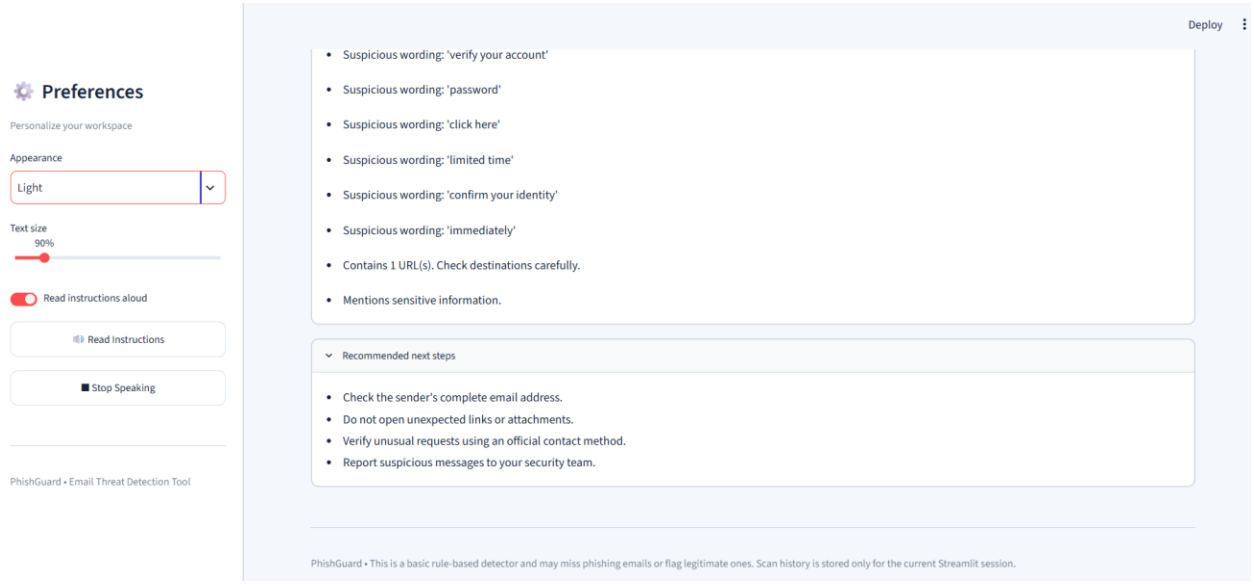


Figure 9 Resulted recommendations

Conclusion

PhishGuard successfully demonstrates a simple, interactive approach to identifying potential phishing emails using rule-based analysis. It detects common warning signs, assigns risk scores, classifies

emails, provides safety recommendations, and allows users to review and export scan history. Its accessible interface helps users understand phishing threats and practice safer email habits.

Limitations**:** The tool relies on predefined rules and may miss sophisticated phishing emails or flag legitimate messages. It does not verify sender identities, scan attachments, or check URLs against external threat databases. Scan history is temporary and is not stored in a permanent database. Therefore, its results should be treated as guidance rather than a guaranteed security verdict.

Future Scope

1. Machine Learning Integration: Implement an ML model to improve phishing email detection and classification.
2. Real-Time URL Checking**:** Integrate threat intelligence services to identify known malicious links.
3. Sender Verification: Analyze sender addresses and email authentication details to identify suspicious sources.
4. Attachment Analysis**:** Add scanning capabilities to detect potentially malicious email attachments.
5. Database Integration: Store scan history securely for long-term access and analysis.
6. Advanced Reporting**:** Generate downloadable PDF reports with detailed findings and recommendations.
7. Improved Detection: Expand detection rules to recognize new phishing techniques and reduce false positives.